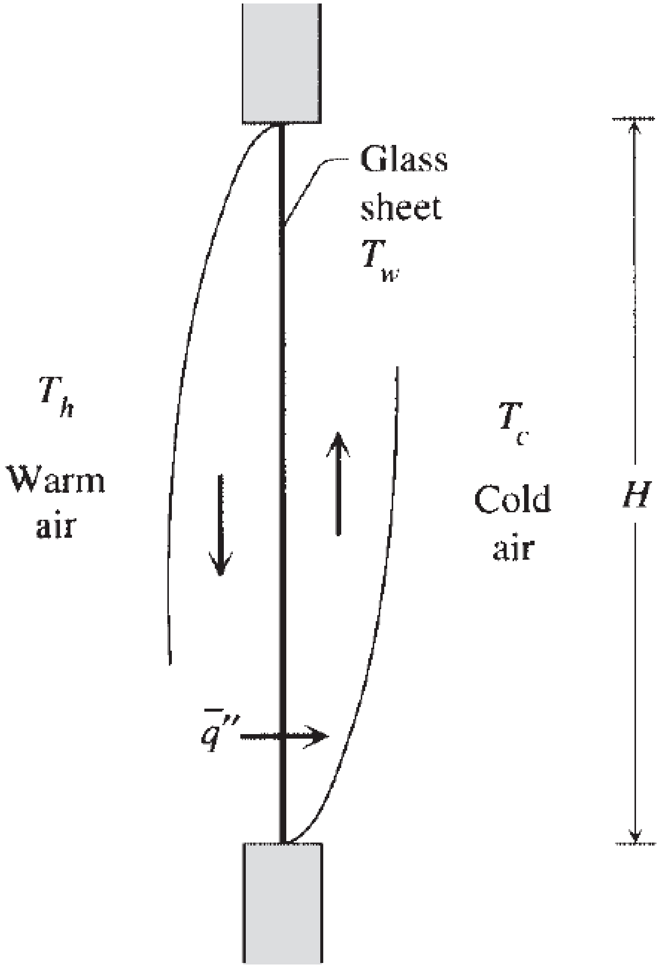
Problem 4

The single-pane window problem consists of estimating the heat transfer rate through the vertical glass layer shown in Fig. The window separates two air reservoirs of temperatures T_h and T_c. Assuming constant properties, laminar boundary layers on both sides of the glass, and a uniform glass temperature T_w. Find the relationship between the average heat flux q'' with T_h to T_c. Expression for the Nu is given as

$$\overline{Nu} \sim 0.517 Ra_H^{1/4}$$

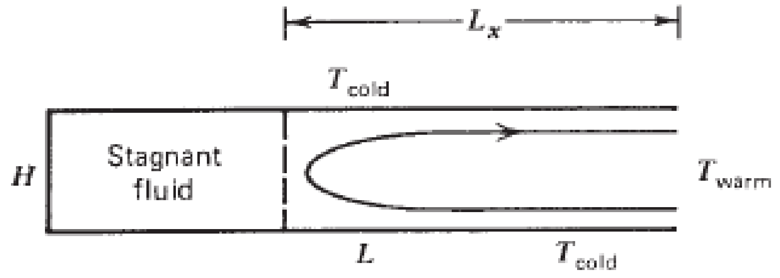


Problem 5

Machined into a solid wall of temperature T is a slender two-dimensional cavity of height H and length L ($H << L$). The cavity communicates laterally with an infinitely large fluid reservoir of temperature $T + T$. The situation is shown schematically in the Figure. Show that if the cavity is slender enough, the buoyancy-driven flow will penetrate the cavity only to a certain depth whose order of magnitude is

$$L_y \sim H \; Ra_H^{1/2}$$

Determine the order of magnitude of the total heat transfer rate between the fluid reservoir and the walls of the cavity.

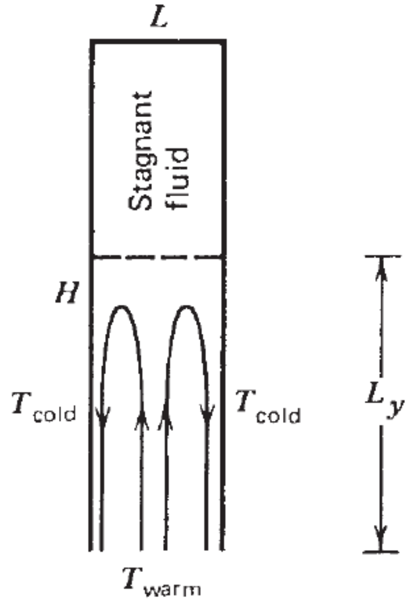


Problem 6

Consider the penetration of natural convection into a vertical slender cavity of height H and gap thickness L (H>>L). As is shown in the Figure, One end of the cavity is closed and the other communicates with a very large fluid reservoir. Rely on scaling arguments to show that when a temperature difference ΔT is established between the fluid reservoir and the walls of the cavity, the flow penetrates vertically to a depth that scales as

$$L_y \sim L\, Ra_L$$

Estimate the order of magnitude of the heat transfer rate between the fluid reservoir and the walls of the cavity, assuming that $L_y < H$.



Vertical penetration